Charlotte Chen

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Research Interests

Computer architecture, with a secondary emphasis on integrated circuits/VLSI. Hardware acceleration and memory-centric architecture: accelerators for ML and data-intensive workloads, processing in/near memory, hardware–software co-design, and AI-assisted, implementation-aware design methodology, grounded in carrying designs from RTL through to silicon.

Education

Columbia University — New York, NY
M.S. in Computer Engineering — *Sept. 2026 – May 2027 (expected)*
B.S. in Computer Engineering | GPA: 3.85/4.00 (Major: 3.86/4.00) — *Sept. 2024 – May 2026*

M.S. Thesis: Agentic microarchitecture design from simulation to implementation (Advisor: Prof. Zishen Wan)

Grinnell College — Grinnell, IA
B.A. in Computer Science, B.A. in Japanese | GPA: 3.96/4.00 (Major: 4.00/4.00) — *Aug. 2021 – May 2024*

Research Experience

Agentic Microarchitecture Design: Do Simulated Gains Survive RTL? — Sept. 2026 – Present
M.S. Thesis Research, COMS 6901 (Advisor: Prof. Zishen Wan) | Columbia University — *New York, NY*

- Studying how often an AI agent's microarchitectural proposals that change cycle behavior but must preserve ISA semantics (e.g., ROB/LSQ sizing, new bypass paths, store-to-load forwarding) keep their simulated gains through RTL implementation, verification, and timing closure.
- Building the evaluation environment on my 2-way out-of-order RISC-V core: porting it to an open Yosys/OpenROAD flow calibrated against Genus/Innovus, measuring baseline PPA and the P&R noise floor, and tiering verification (ISA-model co-simulation, hidden random instruction streams, JasperGold invariant proofs) with mutation testing.
- Developing an `evaluate(patch)`→JSON harness, called by collaborator Christina (Yijing) Jia's agent loop, and a gem5 reference model to compare a prediction-committing agent against simulator-only and RTL-coding agents on performance at achieved frequency per unit area; runs are logged as prediction-vs.-outcome traces to seed a planned self-improvement loop.

Power–Performance Characterization of Heterogeneous Edge SoCs — Jan. 2025 – Oct 2025
Research Assistant (Advisor: Prof. Simha Sethumadhavan) | Columbia University — *New York, NY*

- Studied how dynamic voltage/frequency scaling (DVFS) trades throughput against energy on the NVIDIA Jetson Nano CPU/GPU, using hardware performance counters and fine-grained power measurement to establish energy-delay baselines.
- Built microbenchmarks sweeping compute vs. memory intensity and automated 50+ DVFS runs; separated compute- and bandwidth-bound regimes with roofline analysis.
- Found a perf/W sweet spot where mid-frequency operation outperformed maximum frequency by ∼1.4× by avoiding throttling; presented results at Columbia.

GreenerSKU: Lower-Carbon Datacenter Operations — May 2024 – Aug. 2025
Research Assistant (PIs: Profs. WenKai Guan & Yang Katie Zhao) | Univ. of Minnesota — *Minneapolis, MN*

- Developed a Python datacenter carbon-emission simulator modeling operational carbon under dynamic workload scheduling and variable cooling configurations.
- Extended the framework to embodied carbon, comparing the savings from reusing recycled servers across server-lifespan scenarios.
- Refactored the simulation engine to remove concurrency bottlenecks; validated model outputs against empirical datacenter power data and real-world topologies.

Autonomous Driving Algorithms — Dec. 2023 – Apr. 2024
Research Assistant (PI: Prof. Nuria González-Prelcic) | NC State University — *Raleigh, NC & Remote*

- Developed remote-control hardware and path-planning algorithms for autonomous vehicle control, targeting safety margins and routing efficiency.
- Evaluated time-to-contact (TTC) collision margins and algorithm latency across real-world driving scenarios.

Presentations

C. Chen, A. Peng, "Power and Performance Analysis of GPU under DVFS." Research presentation, Columbia University, Spring 2025. [Slides]

Selected Hardware & Architecture Projects

TSMC 65nm Tape-Out: N-Body Simulation Accelerator | *Verilog, Innovus* Jan. 2026 – Present

- Co-designed (team of 4) a $1.31 \times 1.11\,\text{mm}^2$ accelerator for $O(N^2)$ pairwise force computation, reaching 160M pairs/s at 40 MHz with 90% compute efficiency to enable a real-time 60 Hz interactive 2D simulation.
- Built four parallel 17-stage pipelined two-body cores (1 pair/cycle, 18-cycle latency) with a 5-stage fast inverse square root and 27-bit floating-point intermediates, broadcasting each j-body across cores over 16-body i-tiles.
- Supported an on-chip SRAM mode (5 KiB, $N \leq 512$) and a streaming bypass mode ($N \leq 2048$ at 30 fps) through a 56-pin I/O interface.
- Completed the full RTL-to-GDSII flow with scan-chain DFT, behavioral and post-synthesis verification, and place & route at 67.7% core density; signed off at 40 MHz with 2.30 ns setup slack.
- Co-designed the test PCB; the fabricated chip is now in post-silicon testing.

2-Way Out-of-Order RISC-V Core with SMT | *SystemVerilog, Verilator, Design Compiler* Jan. 2025 – Present

- Designed solo a 2-way superscalar out-of-order RV32IMAF core (R10K-style rename) with 2-thread simultaneous multithreading; synthesized in ASAP7 7nm at ∼660 MHz in 0.05 mm^2, est. ∼70 mW and ∼150 pJ/instruction (pre-layout).
- Built per-thread 32-entry ROBs, a 16-entry shared RS, 2 CDBs, an LSQ with out-of-order loads, store-to-load forwarding and a load-wait predictor (4% over conservative issue), a CVFPU-based FP unit, and LR/SC + AMO atomics.
- Evaluated each microarchitectural feature by its measured gain: a 4-MSHR non-blocking D-cache (1.29× IPC), I-cache stream prefetching (1.15×), and a tournament predictor (mispredicts 12.1% → 10.3%); removed a D-side prefetcher that gained only 0.7%.
- Implemented ICOUNT fetch, per-thread recovery, and thread-hashed indexing (SMT throughput 0.98× → 1.11×); verified per commit against a Spike-checked golden model, with random program pairs and mutation testing.
- Now extending it in a team project (EECS 6693, Fall 2026) that sweeps ROB, issue-queue, register-file, and LSQ sizes through a post-route ASAP7 Genus/Innovus flow, attributing critical paths to microarchitectural structures and testing how well synthesis-level estimates rank configurations against post-route results.

1024-Point Memory-Based FFT Processor in TSMC 65nm | *EECS 6321, Verilog* Jan. 2026 – May 2026

- Designed solo a radix-2 DIF FFT processor around a single complex butterfly PE, four single-port SRAM banks with XOR bank mapping for conflict-free access, and a 512-entry twiddle ROM, using a Q1.15 datapath with per-stage scaling.
- Closed full physical design and signoff (PrimeTime/PrimePower; Calibre LVS/DRC clean) at 250 MHz in 0.47 mm^2 and 2.57 mW, delivering 20.8 MS/s at 123 pJ/sample; 91% of power is in the SRAM banks.
- Post-layout simulation matches the MATLAB golden model bit-exactly over all 10,240 outputs (SQNR 63.0 dB); post-layout HSPICE agrees with PrimeTime within 6% on delay and 4% on power.

TPU-like Accelerator Integration in ESP | *Prof. Luca Carloni; Mentor: Biruk Seyoum* Oct. 2024 – Dec. 2024

- Studied the TPU-like accelerator from the Koios deep-learning benchmark suite (PE microarchitecture, tiling, memory traffic) and integrated it into the ESP SoC through the ESP RTL flow, defining its register map, DMA descriptors, and status interface.
- Built the AXI4-Lite control plane and AXI DMA burst streams for input/weight/partial-sum data; wrote bare-metal and Linux user-space applications to run real DNN workloads on FPGA, validating outputs against Koios golden results.
- Built a generator that produces the accelerator at any PE count, and swept latency against FPGA LUT/DSP utilization to choose per-workload PE counts.

Teaching Experience

Teaching Assistant, Computer Architecture Sept. 2025 – Dec. 2025
Columbia University *New York, NY*

- Supported hardware design labs (Verilog debugging, timing analysis, out-of-order processor simulation) and led discussion sections on pipelining, cache policies, and hazard resolution.

Teaching Assistant, Computer Architecture Jan. 2024 – May 2024
Grinnell College *Grinnell, IA*

- Led weekly lab and discussion sessions on computer architecture fundamentals, assembly programming, and single-cycle datapath execution.

Honors, Grants & Funding

UROP Research Grant ($2,500), University of Minnesota *May 2024*

Funded datacenter carbon-modeling research; co-investigator with Senecy Zhang.

Student Research Fund ($4,000), Grinnell College *Mar. 2023*

Funded Stats2Lab educational software development for the Department of Statistics.

Tau Beta Pi Engineering Honor Society, Columbia University

Dean's List, Columbia University; Grinnell College

Skills

Hardware Description & Verification: SystemVerilog, Verilog, UVM, JasperGold, VCS, Verilator, Verdi, QuestaSim/ModelSim
EDA: Synopsys Design Compiler, PrimeTime/PrimePower, HSPICE; Cadence Genus, Innovus, Virtuoso; Calibre; Yosys/OpenROAD; Vivado
Architectures & Simulators: RISC-V, ESP SoC Platform, Spike, gem5
Programming & Scripting: C/C++, Python, Java, C#, Tcl, Bash
Languages: English, Japanese